Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)

Experiment 2

QUALITITATIVE ANALYSIS OF AMINO ACIDS

Pre-Lab Questions:

- Define the solubility, and activated benzene ring in aromatic amino acids.
- Do all amino acids with aromatic side chain give a positive result in Xanthoproteic test? Why?
- Write the reactions involved in Ninhydrin test.

Precautions:

- In this experiment, you are going to use concentrated nitric acid (in xanthoproteic test section), you have to deal with this strong acid **carefully** because it can cause serious burns if you spell it at your skin.
- Avoid spelling ninhydrin (ninhydrin test section) on your skin since the resulting stain is difficult to wash out. Ninhydrin is a very common chemical used to detect the fingerprints, as the terminal amines or lysine residues in proteins sheds off in fingerprints react with ninhydrin.

Objectives:

- To perform some qualitative tests used to identify amino acids.
- To know the reactions, if exist, that are involved in each test.
- Determine the amino acids that have positive results in each test.
- Be able to calculate the sensitivity limit of ninhydrin test.

Introduction:

Amino acids are the building block of proteins. Proteins are essential components in all organisms and participate in many biological processes such as cell signaling, immune responses, cell adhesion, cell cycle, and others. In general, amino acids composed from an **amine group**, a **carboxylic group**, and a **side chain (R)** that varies among different amino acids and it specifies both the physical properties of proteins like solubility, and the nature of chemical interaction with various compounds (figure 1).

Figure 1. General amino acid structure, amino acids composed from an amine group, a carboxylic group, and a variable side chain (R) https://en.wikipedia.org

Depending on the structure of the side chain, amino acid might be classified as acidic, basic, or neutral amino acids. Amino acids might also be classified to polar (hydrophilic), or non-polar (hydrophobic) according to their side chains. Today, we are going to perform different tests to study different properties of these amino acids. The outcome of those tests depends on the general structure of the amino acid used.

- Solubility tests:

The solubility of amino acids is mainly dependent on the pH of the solvent. The pH changes the structure of the amino acid, for example in acidic solvent both amino and carboxylic groups in amino acid are protonated; in contrast, in basic solvents the aforementioned groups are deprotonated.

Moreover, it could also depend on the polarity of both the amino acid and the solvent used. " like dissolves like". As a result, polar amino acids are dissolved in polar solvents and non-polar amino acids are dissolved in non-polar solvents "like dissolves like"

Since the increase in the number of carbon atoms in the R-chain decreases the polarity of the amino acid, then we expect that the solubility of amino acid to decrease in polar solvents and to increase in non-polar solvent.

Experimental procedure:

Materials needed

- Distilled water (H_2O), hydrochloric acid (HCl), sodium hydroxide (NaOH), acetone (CH_3-CO-CH_3), chloroform ($CHCl_3$).
- Different amino acids (powdered form).

Procedure

- Place a small amount of different powdered amino acids in test tubes
- Add ~ 1 ml of different solvents; H_2O, HCl, NaOH, CH_3-CO-CH_3, and $CHCl_3$.
- Record your results using the words: soluble (+++), slightly soluble (+), and insoluble (-).

- Xanthoproteic test:

Amino acids which contain an **activated** aromatic (benzene) **ring**, form yellow nitro derivatives on heating with concentrated nitric acid, then if a pellet of NaOH is added to the reaction salts of these derivatives are formed and they are orange in color.

Tyrosine, tryptophan, and phenylalanine are three amino acids that have an aromatic ring. Tyrosine and tryptophan have an activated benzene ring; therefore, both of them give yellow color upon performing the xanthoproteic test. On the other hand, the benzene ring in phenylalanine is not activated; as a result it does not undergo nitration with nitric acid **(figure 2).**

Figure 2. Xanthoproteic test: Xanthoproteic test is positive for tyrosine and tryptophan, because they have an activated benzene ring, and negative for phenylalanine, because it has non-activated benzene ring http://nuwanthikakumarasinghe.blogspot.com/2011/05/tests-for-proteins-2.html , access date 01/27/2018.

Experimental procedure:

Materials needed:

- Different solutions of amino acids
- Nitric acid solution
- Pellet of NaOH

Procedure:

- Add about 1 ml of nitric to an equal volume of amino acid solution, warm gently, cool, and observe the color change.
- Add one pellet of NaOH to make the solution strongly alkaline.
- A yellow color in the acid solution, which turns bright orange with NaOH, indicates the positive result.

- Ninhydrin test:

Ninhydrin is a powerful oxidizing agent that reacts with primary and secondary amines. All amino acids reacts with ninhydrin at PH range between (4-8) and produce a colored compound after heating the mixture (Figure 3). The ninhydrin reaction provides a very sensitive test for **detection** of amino acids and in some cases the identification of a specific amino acid.

Figure 3. The Ninhydrin reaction with amino acids: Ninhydrin interacts with amino acids to give a blue-purple color product. Proline and hydroxyproline give yellow-orange color while glutamine and aspargine give a brown color. https://biochemistryisagoodthing.wordpress.com/2013/02/18/ , access date 01/27/2018.

Experimental procedure:

Materials needed:

- Different solutions of amino acids.
- Ninhydrin solution

Procedure

- Place 1 ml of the amino acid solution in a test tube and adjust the PH to about neutrality.

- Add 5 drops of ninhydrin solution and boil for 2 minutes.
- Observe the change in color

- The sensitivity limit of ninhydrin test

Sensitivity limit is defined as the smallest concentration that can be measured by a specific analytical method, in this case the ninhydrin test.

Since the ninhydrin test is used as a very common technique in detecting the fingerprints, then we assume that this test should be very sensitive.

Today we are going to determine the sensitivity limit of ninhydrin test by carrying out the test on serial dilutions of Glycine until negative results is obtained. Next, the sensitivity limit will be calculated according the following equation:

The sensitivity limit= Original concentration X The serial dilution of the last tube that have observed color

Number of serial dilutions= number of dilutions done until the color disappeared

Experimental procedure:

Materials needed:

- Glycine amino acid solution
- Ninhydrin solution

Procedure:

- Make dilutions 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, etc. Look at the flow chart below (Figure 4).
- Calculate the sensitivity limit of the ninhydrin test according to the equation

mentioned above.

Figure 4. A flow chart shows the serial dilution of glycine

Data analysis:

- Write the report of this experiment in your own words.
- In all part, indicate the results according to your work, solubility test, please draw a table and indicate all the solvents and amino acids that you used, and try to justify the results.

References:

- Armbruster, D. A., & Pry, T. (2008). Limit of blank, limit of detection and limit of quantitation. The Clinical Biochemist Reviews, 29(Suppl 1), S49.
- https://www.chemguide.co.uk/organicprops/aminoacids/background.html. 01/27/2018.
- https://medicallaboratorytests.com/xanthoproteic-test-principle-procedure-and-result/, 01/27/2018.
- http://www.chem.ucalgary.ca/courses/351/Carey5th/Ch27/ch27-3-3.html, 01/27/2018

Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 2

Qualitative analysis of amino acids

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Purpose:

Data and calculations:

A: Solubility (soluble: +++, Slightly soluble: +, Insoluble: -)

Solvent// Amino Acid	H_2O	HCl	NaOH	CH_3COCH_3	CH_3CH_2OH	$CHCl_3$
Alanine						
Arginine						
Glycine						

Glutamic acid						

B: Xanthoprotic reaction

Amino acid	Tryptophan	Tyrosine	Alanine	Phenylalanine
Color				

C: Ninhydrin Test:

Amino acid	Tryptophan	Tyrosine	Alanine	Phenylalanine	Proline
Color					

D: Sensitivity Limit for (   ) =

Conclusion:

Errors and problems you encountered and their possible remedies: